Contents

Prerequisites

Technologies for Artificial Intelligence assumes no prior exposure to machine learning. It does assume the background of a first-year MSc student in Computer Science.

What you need

Programming. Solid Python: functions, classes, comprehensions, virtual environments, and comfort reading library documentation. You do **not** need prior experience with NumPy, pandas or scikit-learn — those are introduced in Lessons 1 and 2 — but you should be fluent enough that learning a new library is routine.

Mathematics. Linear algebra (vectors, matrices, matrix products, eigenvalues) and probability (random variables, expectation, variance, conditional probability), at the level of a standard undergraduate Computer Science curriculum. Partial derivatives are used throughout; you should be able to follow a chain-rule argument.

Tools. Git for cloning and pulling the course material. Docker Desktop for the lab environment — no local Python installation is required.

Hardware. A laptop with at least 8 GB of RAM and 20 GB of free disk space. No GPU is needed: every lab runs on CPU.

What you do not need

- Prior machine learning, statistics or data science experience.
- Any paid service. **No API keys, no cloud accounts, no LLM subscriptions.** Every notebook runs offline once the environment is built.
- Your own dataset. All exercises and the final project ship with data.

A note on the mathematics

This course does not hide the mathematics. Handouts contain complete derivations — the normal equation, cross-entropy, the bias-variance decomposition, margin maximisation, backpropagation — because understanding *why* a method works is what separates using a library from doing machine learning.

The lectures themselves stay at the level of results and intuition. The derivations are there for you to work through afterwards, and exam questions will assume you have.

If you are unsure

Work through the Lesson 1 notebooks before the second lesson. If you can follow them comfortably, you are ready. If the Python feels hard rather than merely unfamiliar, speak to the instructor early rather than late.